

Module 08: Transmission Impairments & Signal Characteristics

AKTU BCS603 Focus: Attenuation metrics (dB calculations), Distortion in composite signals, 4 types of noise, and Signal-to-Noise Ratio (SNR).

Transmission Impairments in Physical Media: Attenuation, Distortion & Noise

1. Attenuation (Loss of Energy)

Original Signal

Attenuated Signal

Amp (+dB)

$dB = 10 \log_{10}(P_2 / P_1)$

Negative dB = Loss | Positive dB = Gain

- Signal weakens over distance as heat
- Solved by: Amplifiers / Repeaters
- $dB = 20 \log_{10}(V_2 / V_1)$ for Voltages

2. Distortion (Change in Shape)

In Phase (At Sender)

Out of Phase (At Receiver)

Phase Shift & Delay Spread

Different frequencies travel at different speeds

- Occurs exclusively in composite signals
- Harmonic components arrive at different times
- Creates Inter-Symbol Interference (ISI)

3. Noise (External Corruption)

$SNR = \text{Signal Power} / \text{Noise Power}$

$SNR_{dB} = 10 \log_{10}(SNR)$

- Thermal Noise (Johnson-Nyquist kTB)
- Induced Noise (Motors, power cables)
- Crosstalk (Wire-pair interference)
- Impulse Noise (Spikes/Lightning - data killer)

Key Impairment Formulas & Engineering Solutions

Loss in Decibels: $dB = 10 \log_{10}(P_2 / P_1)$

If $P_2 = 0.5 P_1$, $dB = 10 \log_{10}(0.5) = -3 \text{ dB}$ (Half power loss)

Cascaded Decibels: $dB_{total} = dB_1 + dB_2 + dB_3$

Decibels simply ADD across multi-stage amplifiers and cable spans!

Signal-to-Noise Ratio: $SNR_{dB} = 10 \log_{10}(SNR)$

If $SNR = 1000$, $SNR_{dB} = 10 \log_{10}(1000) = 30 \text{ dB}$ (Clean transmission)

1. Comparative Impairment Diagnostic Matrix

Impairment	Primary Cause	Affected Signals	Engineering Remedy
Attenuation	Medium resistance & thermal absorption over distance	All Analog & Digital signals	Amplifiers (Analog) & Repeaters (Digital)
Distortion	Frequency-dependent propagation velocity (Phase shift)	Composite signals only	Equalizers & delay compensators
Thermal Noise	Random electron agitation (kTB)	All physical media	Cooling; Shielding; Shannon capacity bounds
Crosstalk	Mutual inductance between parallel copper wires	Twisted pair, multi-core cables	Differential signaling & tight wire twisting
Impulse Noise	Lightning strikes, power grid switches, sparking	Digital packets (Burst errors)	Shielding (STP), Error correcting codes (CRC/FEC)

2. Solved AKTU Numerical: Decibel Power Loss

Problem: Ek 10 km optical fiber link par input power $P_1 = 100 \text{ mW}$ hai. Receiver par received power $P_2 = 25 \text{ mW}$ measure hoti hai. Cable ka attenuation dB me aur dB/km me nikalein.

Solution:

$dB = 10 \log_{10}\left(\frac{25}{100}\right) = 10 \log_{10}(0.25) = 10 \times (-0.602) \approx -6.02 \text{ dB}$

Minus sign indicates power loss (Attenuation = 6.02 dB).

Per km attenuation $= \frac{6.02 \text{ dB}}{10 \text{ km}} = 0.602 \text{ dB/km}$.